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Regression Model for Predicting Recyclable Waste Generation

Case Study: Rio Maior, Portugal

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5. Abstract

Efficient management of recyclable waste at the urban scale requires accurate demand prediction at fine spatial resolution, since waste generation and accumulation occur at the container level, and key operational decisions such as container sizing, placement, collection frequency, and routing are made at this scale. This study develops a multivariate linear regression model based on Ordinary Least Squares (OLS) at the **bin level** to explain and predict the intensity of recyclable waste collection demand as a function of local spatial and demographic characteristics. The case study focuses on the municipality of Rio Maior, Portugal, using real operational data from 2020 to 2024.

Raw operational data at the **bin-day** level were first processed under a trip weight allocation assumption, and then aggregated to construct a stable demand indicator for each container, defined as **mean_kg_per_day**. To improve the statistical properties of the dependent variable, a logarithmic transformation was applied using $\log Y = \ln(1 + \text{mean_kg_per_day})$. Explanatory variables include population within a 100 m buffer and land-use share variables within the same buffer, derived through GIS-based spatial analysis. To avoid perfect multicollinearity inherent to compositional land-use shares, one category was excluded and treated as the baseline. Multicollinearity diagnostics based on the Variance Inflation Factor (VIF) indicate values close to unity, confirming the absence of problematic collinearity.

Estimation results show that, in this specification, none of the static spatial variables are statistically significant at conventional levels, except for the intercept. The very low coefficient of determination (R^2) and the non-significant overall F-test indicate that static spatial features alone have limited explanatory power for average daily collection demand. This outcome is consistent with the operational nature of municipal waste systems, where demand variability is largely driven by dynamic and non-spatial factors such as service schedules, collection frequency, day-of-week and seasonal effects, container type, waste stream characteristics, operational policies, and user behavior.

In conclusion, the proposed model should be interpreted as a **transparent and reproducible spatial baseline**, suitable for structuring subsequent analyses and for integration with higher-level decision-support models for container location and collection routing or scheduling. However, it is insufficient on its own to capture the dynamics of daily demand. A logical next step is to shift the unit of analysis to the **bin-day** level and/or to incorporate temporal and operational variables, as well as complementary analyses such as overflow behavior.

Keywords: Recyclable Waste Collection, Bin-Level Demand Modeling, Spatial Regression Analysis, Urban Waste Management, Operational Data Analytics

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6. Introduction

Efficient management of recyclable waste at the urban scale requires accurate and reliable demand forecasts at a fine spatial resolution, as waste generation and accumulation occur at the level of individual collection containers. In practice, key operational decisions such as container capacity selection, container siting, service frequency determination, and collection route design are inherently made at the container level rather than at the neighbourhood or citywide scale. Consequently, aggregated analyses that report only zonal or city-level averages are often insufficient for operational decision support and may result in uneven capacity allocation, increased risk of overflow, and higher collection costs.

The objective of this study is to develop and estimate a **multiple linear regression (MLR)** model at the container level to explain and predict recyclable waste demand as a function of surrounding environmental and spatial characteristics. Within this framework, variables are explicitly defined as a dependent variable (Y) and a set of independent variables (X). The explanatory variables include demographic indicators, land-use composition metrics (expressed as the proportion of each land-use class within a 100-m buffer around each container), and additional spatial attributes derived from Geographic Information Systems (GIS).

The proposed model is developed using real operational data from the municipality of Rio Maior, Portugal, covering the period 2020–2024. Two independent operational data sources are integrated:

- (1) collection operation records from the truck driver's perspective, registered through the collection management system; and
- (2) fill-level sensor data, consisting of time series of container fill percentages. The integration of these data sources enables a distinction between *collected waste* and *accumulated or potentially generated waste*, thereby providing a more comprehensive representation of container-level demand behaviour.

The core assumption underlying this research is that the intensity of recyclable waste generation or accumulation at each container reflects the surrounding pattern of human activity, land-use structure, and relative location within the urban system. In particular, containers located in areas with higher population density or a greater share of commercial, service, or institutional land uses—typically concentrated in central urban areas—are expected to exhibit demand patterns distinct from those located in predominantly residential neighbourhoods. Quantifying these relationships yields two direct practical benefits: first, improving the prediction and interpretation of existing container performance to support service scheduling and capacity allocation; and

second, estimating potential demand at locations without containers to support network expansion and container location decisions.

The regression model developed in this study is not used solely as an independent statistical analysis tool, but rather as an **input module** for subsequent decision-making stages, including container location models (p-median) and collection routing and scheduling models (VRP/IRP). Accordingly, emphasis is placed on data transparency, precise variable definition, and model reproducibility, ensuring that the results are both scientifically defensible and operationally applicable for urban planning and waste management.

The remainder of this section is organized as follows. First, the data sources and data cleaning and standardization procedures are described. Next, the definition of the dependent and independent variables, the rationale for feature selection, and the construction of the final regression dataset are presented. Finally, the regression model specification, diagnostic tests (including multicollinearity control and error assumption checks), and estimation results with managerial interpretation are reported.

7. Conceptual Framework and Regression Methodology

The purpose of the regression analysis in this study is to model the relationship between recyclable waste generation or collection (dependent variable) and a set of demographics, spatial, and land-use explanatory factors (independent variables) at the level of individual urban containers. The output of this regression serves as an input to container location models (p-median) and collection routing and planning models (VRP/IRP).

The primary unit of observation is the **container–day (bin–day)**, meaning that each observation represents the behaviour of a specific container on a specific day. This structure allows the analysis of daily variability and the derivation of stable demand indicators. For the estimation of the baseline regression model, however, daily observations are aggregated to the container level, and a **cross-sectional regression model** is estimated at the bin level.

The statistical model employed is a **classical multiple linear regression estimated using Ordinary Least Squares (OLS)**. The analysis does not involve panel regression techniques, logistic or Poisson models, nor machine-learning-based estimators. Instead, a standard OLS specification is adopted as a baseline explanatory model for spatial demand analysis at the container level.

8. Statistical Evidence Supporting the Use of OLS

The regression outputs exhibit the four canonical components of an OLS model:

estimated coefficients ($\hat{\beta}$), standard errors ($SE(\hat{\beta})$), t-statistics, and p-values. This four-element structure constitutes the definitive statistical signature of an OLS regression. In OLS, the t-statistic is computed as:

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

The explicit presence of these statistics in the model outputs confirms that the estimation method is Ordinary Least Squares. If an alternative modeling framework had been used, different statistical indicators would have appeared. For example:

- Logistic regression models report z-statistics
- Poisson or count models typically report Wald statistics
- Maximum likelihood–based models emphasize log-likelihood values
- Panel data models report variance components such as effects, σ_u, ρ
- The absence of such indicators, combined with the presence of coefficient estimates, standard errors, and t-statistics, unequivocally demonstrates that the model employed in this study is a **cross-sectional OLS regression at the container level**.

this study applies a multiple linear regression model estimated via Ordinary Least Squares as a baseline, interpretable, and spatially explicit tool for analysing recyclable waste demand at the container level. The model is not intended to replace machine learning approaches, but rather to provide a transparent and theoretically grounded foundation that can be directly linked to spatial optimization and operational planning models in subsequent stages of analysis.

9. Regression Computation Process in Five Layers

From raw data ingestion to variable construction, overflow definition, OLS estimation, and result interpretation

5.1. Layer 1. Problem Definition, Unit of Analysis, and Variables

Study Objective

The objective of this analysis is to explain and predict the intensity of recyclable waste collection demand at the level of individual urban containers. The resulting model is designed to serve as a direct input to subsequent decision-support models, including container location planning (p-median) and collection routing and scheduling models (VRP/IRP). The primary emphasis is on developing a transparent, reproducible, and interpretable baseline model that quantitatively links waste collection demand to the surrounding spatial and demographic context of each container.

5.1. Unit of Analysis

Although data are available at multiple temporal and operational levels, the **final regression model is specified as a cross-sectional model at the container (bin-level)**. The data hierarchy is as follows:

- Raw operational level: bin–event
- Intermediate analytical level: bin–day
- Final regression level: bin-level, where each row corresponds to a unique container identifier (idcontentor)

Temporal information (daily or event-based) is used exclusively to construct a stable demand indicator and is subsequently aggregated to the container level. As a result, the final model:

- Is cross-sectional
- Is neither time-series nor panel-based
- Functions as a **baseline spatial regression**

6.1. Definition of Unit of Analysis and Variables in the Regression Model

1) Unit of Analysis

In this study, the raw data are observed at two different levels; however, the **core regression model of the thesis is estimated at the container level (bin-level)**.

Level A: Bin–day (container–day)

At this level, observations are constructed for each container i and each day t . This level is used exclusively to:

- aggregate collection events into daily values,
- derive temporal features such as lags and rolling statistics,
- analyse service dynamics and filling behaviour,
- support exploratory and complementary analyses.

This level **does not generate the final regression coefficients reported in the thesis**.

Level B: Bin-level (container level) – Main model

In the final model:

- Each row of the dataset corresponds to **one container**, uniquely identified by **idcontentor**.
- Daily and event-level data are first aggregated to construct a **stable demand indicator**, which is then transferred to the bin-level.

Accordingly, the main regression model is:

- **cross-sectional,**
- **neither time-series nor panel-based,**
- and serves as a **baseline spatial regression** to quantify the effect of demographic and land-use characteristics on waste collection demand intensity.

5.1. Model Specifications

To avoid any ambiguity regarding data structure and coefficient interpretation, two model specifications are explicitly distinguished.

1.1.1 Specification A (complementary): Bin–day model

- Purpose: analysis of temporal patterns, lag effects, service intervals, and behavioural dynamics.
- Role: supplementary and exploratory.

2.1.1 Specification B (main model): Bin-level regression

- Purpose: construction of a transparent, reproducible baseline model suitable for spatial decision-making and integration with p-median, VRP, and IRP frameworks.

- **All regression results reported in the Excel file correspond exclusively to this specification.**

6.1. Dependent Variable (Y)

The dependent variable represents a stable indicator of waste collection demand at the container level and is defined.

3.1.1 Conceptual clarification

Two operational data sources are available: **driver records** and **container sensors**. While both allow the definition of dependent variables, the key point is that:

In the final Excel-based regression (Specification B), the dependent variable is derived solely from driver data.

Sensor data are not used as the primary dependent variable in the main model and serve only a complementary analytical role.

7.1. Dependent Variable in Specification B (Main Model)

(B1) Stable demand indicator at bin-level

For each container i , the raw dependent variable is defined as:

$$Y_i = mean_kg_per_day_i$$

where:

- $mean_kg_per_day_i$: represents the **average daily collected weight** from container i over the entire observation period, expressed in **kg/day**.

Formally:

$$mean_kg_per_day_i = \frac{1}{T_i} \sum_{t=1}^{T_i} kg_day_{i,t}$$

with:

- $kg_day_{i,t}$: total weight collected from container i on day t, after aggregating all service events occurring on that day;
- T_i : number of days with valid operational data for container i.

This indicator smooths short-term fluctuations and provides a **robust proxy for long-term demand intensity** at the container level.

(B2) Regression form of the dependent variable

Due to the strong right-skewness typically observed in waste collection data, a logarithmic transformation is applied.

If all values are strictly positive:

$$\log Y_i = \ln (mean_kg_per_day_i)$$

If zero values may occur:

$$\log Y_i = \ln (1 + mean_kg_per_day_i)$$

The exact transformation applied must be stated explicitly in the final report, consistent with the Excel implementation.

3.3 Complementary dependent variables (Specification A only)

For completeness, but not used in the main regression:

- **Driver-based daily variable:**

$$Y_{i,t}^{driver} = kg_day_{i,t}$$

- **Sensor-based daily variable:**

$$Y_{i,t}^{sensor} = \max (Enchimento_{i,t})$$

These variables are used solely for exploratory and behavioural analyses.

8.1. **Independent Variables (X)**

Independent variables consist primarily of static spatial and demographic characteristics within a 100-meter buffer around each container, including:

4.1.1 **General principle**

Under **Specification B**, all independent variables are:

- **static,**
- **spatial,**
- defined at the **bin-level.**

They are primarily extracted from GIS layers using a **100-metre buffer** around each container.

5.1.1 **Independent Variables Used in the Main Model**

(X1) Demographic variable

$$Pop_{100m_i}$$

Population residing within a 100-metre radius of container i.

(X2) Land-use composition variables

For each container i, the share of **land-use** class k within the 100-metre buffer is defined as:

$$Pct_{k_i}$$

The following land-use classes are included:

- $Pct_RESIDENTIAL_i$
- $Pct_COMMERCIAL_i$
- $Pct_INDUSTRIAL_i$
- $Pct_INSTITUTIONAL_i$
- $Pct_ADMINISTRATIVE_i$
- $Pct_HEALTHCARE_i$
- $Pct_CULTURAL_i$
- $Pct_RECREATIONAL_i$

- $Pct_TRANSPORT_i$
- Pct_OTHER_i

Additional classes are included where present in the GIS dataset.

6.1.1 Baseline rule and multicollinearity control

Since land-use percentages typically sum to 100%:

$$\sum_k Pct_k_i = 100$$

one category must be excluded to avoid perfect multicollinearity.

In this study, Pct_OTHER_i is omitted and acts as the **baseline category**.

All estimated coefficients therefore represent marginal effects relative to this omitted class.

7.1.1 Regression Model (Specification B)

The estimated cross-sectional log-linear model is:

$$\log Y_i = \beta_0 + \beta_1 Pop_100m_i + \sum_{k \in K} \beta_k Pct_k_i + \varepsilon_i$$

where ε_i is the error term.

In the final regression model (Specification B), the dependent variable $\log Y_i$ is constructed from driver data and is rooted in the container-level indicator $mean_kg_per_day_i$. The independent variables consist of Pop_100m_i and land-use composition variables Pct_k_i defined within a 100-metre buffer around each container. To control for perfect multicollinearity, the category Pct_OTHER_i is omitted and serves as the baseline. Sensor data are not used as the primary dependent variable in this model and are employed only as complementary information for analysing filling behaviour and supporting robustness checks.

6.1. Layer 2. Data Ingestion and Construction of the Base Dataset

8.1.1 Input Files and Their Roles in the Processing Chain

This study relies on four primary datasets, each with a clearly defined role in the data-processing pipeline:

1. **Enchimentos_com_Recolhas_RioMaior_2023.csv**

Raw collection operation data recorded by drivers and the operational system. This dataset forms the basis for event-level construction and extraction of

collected waste quantities.

2. **Enchimentos_de_Sensores_RioMaior_2023.csv**

Raw time-series data from container fill-level sensors. These data are used for quality control, overflow analysis, and linking near-emptying fill conditions to collection events.

3. **bins_spatial_features_binlevel.csv**

Spatial and land-use attributes surrounding each container at the bin level. This file is the primary source of Pop_100m and Pct_* variables.

4. **regdata_input.csv**

The final regression-ready dataset, where each row represents a single container and includes the dependent variable and all explanatory variables.

9.1.1 Intermediate Outputs

Several intermediate datasets are generated during processing:

- **event_stops_merged.csv**: event-level data after merging driver and sensor information
- **trip_summary.csv**: trip-level summaries
- **cont_daily.csv**: bin-day indicators
- **cont_stats.csv**: bin-level summary including mean_kg_per_day

10.1.1 Integration of Operational Data Sources

Driver and sensor datasets are complementary. Driver data provide the primary measure of collected quantities, while sensor data offer a more accurate representation of actual fill behavior and overflow conditions. These datasets are integrated using:

- **Common identifier**: idcontentor
- **Temporal rule**: linking the last sensor reading prior to the emptying time (backward temporal matching)

7.1. Data Sources and Construction of the Regression Base Dataset

According to the data structure, this study relies on **two primary operational data sources** and one spatial (GIS) data source. These datasets are linked using common identifiers and explicit temporal rules in order to construct a consistent dataset at the **bin-event** level and, ultimately, to derive a stable demand indicator at the **bin level**.

11.1.1 Collection Operations Data (Driver Data)

Enchimentos_com_Recolhas_RioMaior_2023.csv

This file contains raw collection operation records reported by truck drivers and the operational management system. It represents the **primary source for estimating collected waste quantities**.

Key Columns

- **idcontentor**: Unique container identifier (primary key for data integration)
- **idrecolha**: Collection trip or event identifier
- **Data de início + Tempo de início**: Start time of container emptying
- **Peso totl**: Total collected weight per trip (kg)
- **Km totais**: Total distance traveled during the trip (km)
- **Enchimento**: Fill level (%) reported by the driver prior to emptying

8.1. Methodological Note

The weight reported in this dataset is recorded at the **trip level**, not necessarily at the individual container level. Therefore, a **weight allocation assumption** is required when analyzing demand at the container level. This assumption is explicitly defined and discussed in later stages of the methodology.

After data cleaning, time standardization, and consistency checks, this dataset is transformed into the **event_stops** table, where **each row represents one container serviced during one collection trip**.

Structure of the Driver Dataset

Column	Description
idcontentor	Container identifier (join key)
Tipo de contentor	Container type (AMBI, OVO, VRL, etc.)
Volume do tipo de contentor	Container volume (liters)
description	Waste type (paper, glass, plastic, etc.)
Longitude / Latitude	Container coordinates
Data de início	Date of emptying start
Tempo de início	Time of emptying start
Data de fim	Date of emptying end
Tempo de fim	Time of emptying end
Enchimento	Fill level (%)
idrecolha	Collection trip/truck identifier
Peso totl	Collected weight (kg)
Km totais	Distance traveled (km)

12.1.1 Container Fill-Level Sensor Data

Enchimentos_de_Sensores_RioMaior_2023.csv

This dataset contains **time-series sensor readings of container fill levels** and serves as a complementary source to the driver-reported data. Sensor data are used to observe actual accumulation behavior, identify overflow events, and validate fill levels close to collection times.

Key Columns

- **idcontentor**: Unique container identifier
- **Data da leitura + Tempo de leitura**: Sensor reading timestamp
- **Enchimento (%)**: Fill level measured by the sensor

Main Uses of Sensor Data

- Analysis of real container filling behavior
- Definition and validation of overflow indicators
- Temporal alignment of fill conditions immediately prior to emptying

Temporal Matching of Sensor and Driver Data

Using **backward temporal matching** (analogous to `merge_asof`), the **last sensor reading prior to each emptying event** is linked to the corresponding collection record. The output of this step is the **event_stops_merged** dataset, in which both driver-reported and sensor-based fill levels are available for each collection event when sensor data exist.

Structure of the Sensor Dataset

Column	Description
idcontentor	Container identifier (join key)
Tipo de contentor	Container type
Volume do tipo de contentor	Container volume (liters)
description	Waste type
Longitude / Latitude	Container coordinates
Data da leitura	Reading date
Tempo de leitura	Reading time
Enchimento	Fill level (%)
idrecolha	Trip identifier (if recorded)
Peso totl	Collected weight (if recorded)
Km totais	Distance traveled (if recorded)

13.1.1 Spatial and Land-Use Data (GIS)

bins_spatial_features_binlevel.csv

This file stores **static spatial attributes of each container** and constitutes the main source of explanatory variables in the regression model. All features are extracted

within a **100-meter buffer** around each container using GIS processing.

Main Contents

- Surrounding population
- Land-use composition
- Distance to city business district (CBD)
- Physical container attributes (volume, container type, waste type) to understand stream Material

Key Derived Variables

- **Pop_100m**: Population within a 100-meter buffer
- **Land-use shares**

14.1.1 Layer 3. Data Quality Control and Error Management (Including Overflow)

At this stage, it is essential to clearly distinguish between two fundamentally different concepts:

- **Operational overflow (container overflow)**: a capacity and service-level issue based on fill percentage
- **Computational overflow**: a numerical processing issue resulting in ∞ or NaN values

15.1.1 Data Cleaning and Harmonization

- Files are read using appropriate encodings (UTF-8 or Latin-1)
- Column names are standardized across the entire pipeline
- idcontentor is converted to a uniform data type (string)
- Invalid records are removed or corrected, including:
 - Negative fill levels
 - Negative weights
 - Inconsistent timestamps (end time preceding start time)
 - Unrealistic or out-of-bound spatial coordinates

3.2 Construction of Standardized Datetime Variables

For both driver and sensor data:

- Date and time fields are combined into standardized datetime variables
- Records are temporally sorted to ensure correct sensor–event alignment

3.3 Definition of Operational Overflow

Operational overflow is defined as follows:

- $\text{overflow_driver} = 1$ if $\text{fill_driver} \geq 100$, otherwise 0
- $\text{overflow_sensor} = 1$ if $\text{fill_sensor} \geq 100$, otherwise 0
- $\text{overflow_combined} = \max(\text{overflow_driver}, \text{overflow_sensor})$

Methodological note: overflow observations are **not removed**. They are labeled to:

- Avoid sample-selection bias
- Preserve information for supplementary analyses
- Enable managerial interpretation related to capacity shortages or service delays

3.4 Computational Overflow and Mitigation Strategies

Computational overflow occurs when numerical values exceed computational limits and become ∞ or NaN. Common causes include:

- $\ln(0)$ or $\ln(\text{negative values})$
- High-order powers or multiplication of large variables
- Unit inconsistencies (e.g., km vs. m)

Standard mitigation strategies include:

- Logarithmic transformations using $+1$ or ε
- Standardization or normalization of large-magnitude variables
- Unit consistency checks prior to data integration
- Outlier labeling and sensitivity analysis instead of blind removal

Layer 4. Regression Computation (From Events to Bin-Level OLS)

4.1 Construction of the Event-Level Table (bin–event)

Each row represents the servicing of one container during a collection event and includes:

- idrecolha , idcontentor
- datetime_start
- fill_driver
- Peso_totl and Km_totais (typically recorded at the trip level and repeated across stops)

4.2 Temporal Matching of Sensor and Driver Events

For each emptying event:

- Key: idcontentor

- Rule: last sensor reading prior to datetime_start
- Maximum tolerance window: 2 hours
- Method: backward temporal matching (merge_asof-like)

The output is **event_stops_merged.csv**. Missing sensor data for some events are accepted and treated as missing, not as errors.

4.3 Key Assumption: Allocation of Trip Weight to Containers

Since collected weight is recorded at the trip level and individual stop weights are unavailable, an equal allocation assumption is applied:

$$Peso_{bin} = \frac{Peso_{trip}}{n_{bins}}$$

This assumption is common in operational studies; however, its limitations are explicitly acknowledged in the Limitations section.

4.4 Computation of Daily Demand (bin-day)

After weight allocation, data are aggregated at the (idcontentor, date) level:

- kg_day
- mean_fill_driver_day
- mean_fill_sensor_day
- overflow_driver_day, overflow_sensor_day

Output: **cont_daily.csv**

4.5 Construction of the Final Dependent Variable

For each container i:

$$\text{mean_kg_per_day}_i = \frac{1}{T_i} \sum_t kg_{i,t}$$

where T_i is the number of valid observation days. Output: **cont_stats.csv**

4.6 Final Regression Dataset

The following datasets are joined by idcontentor:

- cont_stats.csv
- bins_spatial_features_binlevel.csv

Final output: **regdata_input.csv**

OLS Regression Model at the Bin Level

11.1 Unit of Analysis and Dependent Variable

The model is estimated as a cross-sectional OLS regression at the **bin level**, meaning that each observation represents a unique bin ($N = 452$). The daily **bin-day** records are used only to construct a stable demand indicator at the bin level.

The dependent variable is defined as:

$$\log Y_i = \ln(1 + \text{mean_kg_per_day}_i)$$

This transformation ($\log 1p$) is applied to reduce right skewness, limit the influence of outliers, and promote a more stable error variance.

11.2 Explanatory Variables and the Baseline Category for Land-Use Shares

The explanatory variables include:

- **Pop_100m_i**: population within a 100 m buffer
- **Pct_{k,i}**: land-use share of class k within same buffer (fractions between 0 and 1)

Because land-use shares are compositional (their sum is approximately 1), one class must be omitted to avoid perfect multicollinearity. In this project:

- **Pct_OTHER** is excluded and treated as the **baseline**.

Accordingly, the coefficients of **Pct_*** represent **effects relative to the baseline category (OTHER)**, holding the other variables constant.

12) Model Form and Estimated Equation

12.1 General Model Form

$$\log Y_i = \beta_0 + \beta_{Pop} Pop_{100m_i} + \sum_k \beta_k Pct_{k,i} + \varepsilon_i$$

12.2 Estimated Equation (as reported in regression_coefficients.csv)

$$\begin{aligned} \hat{\log Y} = & 5.275631 + 0.344192 Pct_RESIDENTIAL + 0.514317 Pct_INSTITUTIONAL \\ & + 1.199277 Pct_CULTURAL - 0.559110 Pct_TRANSPORT \\ & - 6.374 \times 10^{-7} Pop_{100m} + 0.397734 Pct_HEALTHCARE \\ & - 0.680315 Pct_ADMINISTRATIVE + 0.080507 Pct_COMMERCIAL \\ & + 0.048440 Pct_INDUSTRIAL + 0.022054 Pct_RECREATIONAL \end{aligned}$$

12.3 Back-Transformation to the Original Scale (kg/day)

To convert predictions from the log scale back to the original demand scale:

$$\text{mean_kg_per_day} = \exp(\log Y) - 1$$

If the goal is unbiased prediction of the conditional mean on the original scale, direct exponentiation can introduce retransformation bias. If needed, a **smearing estimator** (Duan) can be applied in the prediction stage.

13) Coefficients and Statistical Inference (Coefficients)

The OLS coefficients are extracted from **regression_coefficients.csv**.

Variable	Coefficient (β)	Std. Err	t	p-value	Significance
Intercept	5.275631	0.582109	9.0630	5.50e-18	***
Pct_RESIDENTIAL	0.344192	0.730749	0.4711	0.6378	
Pct_INSTITUTIONAL	0.514317	0.912158	0.5638	0.5732	
Pct_CULTURAL	1.199277	2.447939	0.4898	0.6246	
Pct_TRANSPORT	-0.559110	1.128226	-0.4956	0.6204	
Pop_100m	-6.374e-07	1.942e-06	-0.3282	0.7429	
Pct_HEALTHCARE	0.397734	0.763129	0.5211	0.6025	
Pct_ADMINISTRATIVE	-0.680315	1.657369	-0.4105	0.6816	
Pct_COMMERCIAL	0.080507	0.233700	0.3445	0.7306	
Pct_INDUSTRIAL	0.048440	0.129119	0.3751	0.7078	
Pct_RECREATIONAL	0.022054	0.129994	0.1697	0.8654	

Direct statistical interpretation: In this model run, with the exception of the intercept, none of the explanatory variables is statistically significant even at the 0.10 level. Therefore, the current specification should be treated primarily as a **spatial baseline** rather than a strong inferential model for identifying independent effects of land use or buffered population.

Technical note: A statistically significant intercept alone typically indicates the existence of a baseline level of $\log Y$, while the selected spatial predictors are not capturing the variation in demand across bins.

14) Model Fit

According to **regression_metrics.csv**:

- Number of observations: N=452
- R2=0.0130

- Adj.R2=-0.0100
- AIC = 1349.476
- BIC = 1398.885
- F-test p-value = 0.9504

Methodological interpretation: The very low R2 and the non-significant overall F-test indicate that static spatial predictors (land-use shares and buffered population) have limited explanatory power for average daily collection demand. In practice, most demand variability in municipal service systems is often driven by non-spatial and dynamic factors, such as:

- operational planning and service policy (collection frequency, route allocation, operational decisions)
- temporal effects (day of week, month, season, holidays, short-term events)
- equipment and material-flow characteristics (bin type, capacity, waste stream, operational constraints)
- user behavior and local participation patterns

A defensible thesis conclusion is that a purely static spatial model is insufficient as the primary explanation of daily demand, and that explanatory performance requires either (i) moving to the **bin-day** level or (ii) incorporating temporal and operational covariates.

15) Multicollinearity (VIF) and Detection of Strong Correlation

Multicollinearity among the explanatory variables was assessed using the **Variance Inflation Factor (VIF)**. VIF measures how much the variance of an estimated regression coefficient is inflated due to linear dependence with other predictors.

A commonly accepted **rule of thumb** for interpreting VIF values is as follows:

- **VIF < 5** → Excellent; no multicollinearity concerns
- **5 ≤ VIF < 10** → Acceptable; moderate multicollinearity, warrants caution
- **VIF ≥ 10** → Problematic; severe multicollinearity, variables should be removed, combined, or redefined

In this study, all VIF values are close to 1, indicating that the explanatory variables are largely independent and that multicollinearity is not a concern. Therefore, any lack of statistical significance in the estimated coefficients is not attributable to collinearity issues, but rather to the limited explanatory power of the selected predictors.

Based on **vif_table.csv**, all VIF values are close to 1 (approximately 1.00 to 1.06).

Conclusion: There is no evidence of problematic multicollinearity. The lack of explanatory power is not driven by collinearity, but rather by the limited scope of predictors restricted to static spatial variables.

16) Direction-of-Effect Interpretation (Descriptive Only)

Since **Pct_*** variables are fractions between 0 and 1, an increase of 0.10 corresponds to a **10 percentage-point** increase in the land-use share within the buffer.

With the log1p specification, if the predicted change in $\log Y$ is Δ , the relative change in $(1+Y)$ is approximately:

$$\frac{1 + Y_{new}}{1 + Y_{old}} \approx e^{\Delta}$$

Descriptive example: Cultural land use

If **Pct_CULTURAL** increases by 0.10:

$$\Delta \log Y \approx 1.199277 \times 0.10 = 0.1199$$

$$e^{0.1199} \approx 1.127$$

This implies an approximate **12.7% increase** in $(1+Y)$.

Important: because the p-value is high, this is strictly a **directional/descriptive** interpretation and should not be reported as an inferential conclusion.

Descriptive example: Population

An increase of 10,000 people in **Pop_100m**:

$$\Delta \log Y = (-6.374 \times 10^{-7}) \times 10000 = -0.00637$$

$$e^{-0.00637} \approx 0.9936$$

This corresponds to about a **0.6% decrease** in $(1+Y)$. This effect is also non-significant and remains descriptive.

17) Short Summary Suitable for Emailing a supervisor

- Daily bin–day records were aggregated to a stable bin-level demand indicator (**mean_kg_per_day**, in **cont_stats.csv**).
- Spatial predictors within a 100 m buffer were extracted (**Pop_100m** and **Pct_***, in **bins_spatial_features_binlevel.csv**).
- A final regression dataset was built with 452 bins (**regdata_input.csv**).
- The dependent variable was defined as $\log Y = \ln(1 + \text{mean_kg_per_day})$
- An OLS model was estimated and outputs were reported (**regression_coefficients.csv**, **regression_metrics.csv**).

Key result: Static spatial predictors alone provide very limited explanatory power. A logical next step is to move to the **bin–day** level and/or add temporal and operational variables (day-of-week, month/season, number of services, time since last service, bin type, waste stream, sensor indicators, routing/fleet features).

9.1. Logarithmic Transformation of the Dependent Variable and Model Diagnostics

Prior to estimating the regression model, a logarithmic transformation was applied to the dependent variable. This procedure represents a standard methodological practice in regression analysis and is typically adopted when the dependent variable exhibits strong right-skewness or when its variance increases with the mean (heteroskedasticity). The dependent variable was logarithmically transformed prior to model estimation to reduce skewness and stabilize the error variance.

In this study, the dependent variable is defined as:

$$\log Y = \ln (\text{mean_kg_per_day})$$

The application of this transformation serves several purposes:

- to reduce right-skewness in the distribution of the dependent variable,
- to mitigate the influence of extreme observations (outliers),
- to stabilize the variance of the error term,
- and to improve compliance with the classical assumptions of linear regression.

The final regression model is therefore estimated in the logarithmic scale. To obtain predictions in the original measurement units (kilograms per day), the inverse exponential transformation is applied:

$$\hat{\text{mean_kg_per_day}} = \exp (\hat{\log Y})$$

It should be noted that when the objective is accurate prediction on the original scale, the exponential back-transformation may introduce bias. In such cases, bias-correction techniques such as the **Smearing Estimator** may be employed. However, given that the primary focus of this analysis is on structural interpretation and comparative model assessment rather than point forecasting, the standard exponential transformation is considered sufficient.

Additional Diagnostic Considerations

In addition to the logarithmic transformation, several diagnostic checks were conducted at a descriptive level to assess model adequacy:

- examination of residual patterns to detect systematic structure,
- qualitative assessment of heteroskedasticity,
- sensitivity analysis with respect to outliers through identification, labeling, and comparison of model behavior.

These diagnostics indicate that the logarithmic transformation of the dependent variable improves coefficient stability and overall model behavior, supporting its use in the final model specification.

10.1. **Variables Not Included in the Current Regression Specification**

Several potentially relevant explanatory factors were not incorporated into the current regression model. Their exclusion is deliberate and reflects data scope and model parsimony considerations rather than methodological oversight.

Overflow events were not included in the regression analysis. However, overflow can be systematically derived from sensor data using standard definitions based on sustained high fill levels beyond operational thresholds. As such, overflow is well suited to be analyzed as a complementary outcome variable or modeled separately using a dedicated specification focused on service reliability and risk conditions.

Street network density was not considered in the regression model. Although network density may influence accessibility and collection efficiency, it represents an infrastructural characteristic that is more directly related to routing and operational performance than to baseline demand generation at the bin level.

Distance to the city center was also excluded from the regression. While this variable may capture aspects of urban hierarchy and land-use intensity, its effect is partially embedded in land-use composition and population distribution variables already considered, and its explicit inclusion was deferred to avoid redundancy in the baseline spatial specification.

Rainfall and weather conditions were not incorporated into the regression model. Precipitation primarily affects short-term waste generation behavior and collection operations and is therefore more appropriately analyzed in a temporal or bin-day framework rather than in a cross-sectional bin-level model.

The current regression is intentionally specified as a **baseline spatial model** using static neighborhood characteristics only. Dynamic, operational, and environmental variables such as overflow, routing constraints, network structure, distance to city center, and rainfall are more suitably addressed in extended specifications or separate analyses focused on temporal dynamics, operational performance, or service reliability.

- For the construction of the final regression dataset, a **cross-sectional bin-level unit of analysis** was adopted, meaning that each row represents a single bin with a unique identifier. The raw operational data, originally recorded at the trip and bin–day levels, were first processed using a **weight allocation assumption**, whereby the collected trip weight was distributed among the bins serviced during that trip. Subsequently, a **daily demand per bin** was computed and then aggregated to derive a stable demand indicator at the bin level. Accordingly, although the original information exists at daily or inter-emptying intervals, the final regression model is estimated on **bin-level aggregated data**, rather than on bin–day observations or inter-collection intervals.

11.1. Conclusion

In this study, a multivariate linear regression model based on Ordinary Least Squares (OLS) was developed at the **bin-level** to explain the intensity of recyclable waste collection demand using spatial characteristics around each container. Daily operational data recorded at the **bin–day** level was first aggregated to construct a stable demand indicator, **mean_kg_per_day**, for each bin. To improve the statistical properties of the dependent variable, a logarithmic transformation $\log Y = \ln(1 + \text{mean_kg_per_day})$ was applied.

The results indicate that, in the current specification, **static spatial variables**—namely population within a 100 m buffer and land-use shares—have **very limited explanatory power** for variations in average daily collection demand. This is reflected in the lack of statistical significance of explanatory variables (except for the intercept), the very low R^2 , and the non-significant overall F-test. Variance Inflation Factor (VIF) diagnostics confirm that multicollinearity is not an issue; therefore, the weak performance of the model is attributable to the intrinsic limitations of static spatial predictors rather than statistical instability.

From a methodological perspective, these findings are consistent with the nature of urban service systems. A substantial share of demand variability in waste collection is typically driven by **temporal and operational factors**, such as service frequency, routing decisions, day-of-week and seasonal effects, container characteristics, waste type, operational policies, and user behaviour—factors not included in the present spatial-only specification.

the proposed model should be interpreted as a **transparent and reproducible spatial baseline** rather than a fully explanatory demand model. It is suitable for structuring subsequent analyses and for integration with higher-level decision-support models, but it is insufficient on its own to capture daily demand dynamics. A logical next step is to **shift the unit of analysis to the bin–day level** and/or to **incorporate temporal**,

operational, and behavioural variables, potentially complemented by separate analyses of overflow and service performance indicators.